



Python Programming Intern

Name: Amima Batool

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Project no. : Third

Project name: Random Password Generator

Random Password Generator

Introduction:

Random Password Generator is a simple command-line application developed using Python. The program generates a random password according to the length entered by the user. It uses uppercase and lowercase letters, numbers, and special characters to create a random combination of characters. This project was developed as part of the Python Programming Internship at DecodeLabs.

Tools & Technologies Used:

- VS Code
- Python
- GitHub

Objectives of the Project:

The main objectives of the Random Password Generator are:

- To develop a simple password generator using Python.
- To practice Python programming fundamentals.
- To learn how to use Python's built-in random and string modules.
- To generate passwords of different lengths according to user input.
- To understand the use of loops, variables, and functions/modules in a practical project.
- To gain hands-on experience in developing a command-line application.

Features:

Following are the features of the Random Password Generator:

1. Allows the user to enter the desired password length.
2. Generates a random password automatically.
3. Uses uppercase and lowercase letters.
4. Uses numbers from 0–9.
5. Uses special characters and punctuation marks.
6. Combines different types of characters to generate a random password.
7. Displays the generated password on the terminal.
8. Simple and easy-to-use command-line interface.

Python Concepts Used:

Following fundamental Python Concepts are used in this project:

➤ Variables:

```
print()
length = int(input("Enter Length: "))
print()
```

```
chars = string.ascii_letters + string.digits + string.punctuation
password = ""
```

➤ User input using input():

```
print()
length = int(input("Enter Length: "))
print()
```

➤ Type conversion using int():

```
print()
length = int(input("Enter Length: "))
print()
```

➤ for loop:

```
for i in range(length):
    password += random.choice(chars)

print("Random Password: ", password)
print()
print("---Password Generated Successfully!---")
print()
```

➤ String concatenation:

```
chars = string.ascii_letters + string.digits + string.punctuation
```

➤ random module and string module:

```
import random
import string
```

➤ random.choice():

```
for i in range(length):
    password += random.choice(chars)
```

➤ Combination of character:

```
print("Words used to generate password: ")
print(string.ascii_letters)
print(string.digits)
print(string.punctuation)

print()
length = int(input("Enter Length: "))
print()

chars = string.ascii_letters + string.digits + string.punctuation
```

➤ Printing output using print():

```
print("Random Password: ", password)
print()
```

Working of the Program:

The execution process of the program is as follows:

1. The random and string modules are imported.

```
#####Random Password Generator#####  
import random  
import string
```

2. The program displays the title of the Random Password Generator and also displays the types of characters that can be used to generate the password.

```
print("-----Random Password Generator-----")  
print()  
  
print("Words used to generate password: ")  
print(string.ascii_letters)  
print(string.digits)  
print(string.punctuation)
```

3. Letters, Digits and Punctuation/special characters will be displayed.

```
Words used to generate password:  
abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ  
0123456789  
!"#$%&'()*+,-./:;<=>?@[\]^_`{|}~  
  
Enter Length: 
```

4. The user is asked to enter the desired password length and the program combines letters, digits, and punctuation into one character set.

```
print()  
length = int(input("Enter Length: "))  
print()  
  
chars = string.ascii_letters + string.digits + string.punctuation
```

```
Enter Length: 8
```

```
Random Password: i=JN>1%S
```

```
---Password Generated Successfully!---
```

5. A for loop runs according to the length entered by the user. During each loop, random.choice() selects one random character from the character set.

```
for i in range(length):  
    password += random.choice(chars)  
  
print("Random Password: ", password)  
print()
```

6. The selected characters are added together to form the password.

```
s = string.ascii_letters + string.digits + string.punctuation
```

7. Finally, the generated random password is displayed on the terminal.

```
Random Password: i=JN>1%S
```

Application Flow:

The overall basic workflow of the application is as follow:

Start → Display “Random Password Generator” → Display available characters → Ask user to enter password length → create a combined character set → random.choice() → Repeat until required length is reached → Display Generated Password → END.

Code:

```
#=====Random Password Generator=====
```

```
import random
import string
```

```
print("-----Random Password Generator-----")
print()
```

```
print("Words used to generate password: ")
print(string.ascii_letters)
print(string.digits)
print(string.punctuation)
```

```
print()
length = int(input("Enter Length: "))
print()
```

```
chars = string.ascii_letters + string.digits + string.punctuation
```

```
password = ""
```

```
for i in range(length):
    password += random.choice(chars)
```

```
print("Random Password: ", password)
print()
print("---Password Generated Successfully!---")
print()
```

Sample Output:

```
PS C:\Users\amima\OneDrive\Desktop\VS CODE folder> python Random_Password_Generator.py
-----Random Password Generator-----

Words used to generate password:
abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ
0123456789
!"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~

Enter Length: 8

Random Password: i=JN>1%S

---Password Generated Successfully!---
```

Testing:

The application was tested by performing each operation:

Sr.	Task	Result	Status
1.	Entered length “8”	Password Generated Successfully!	Passed
2.	Entered length “10”	Password Generated Successfully!	Passed
3.	Entered length “15”	Password Generated Successfully!	Passed
4.	Entered length “7”	Password Generated Successfully!	Passed
5.	Entered length “90”	Password Generated Successfully!.	Passed

Conclusion:

The Random Password Generator successfully demonstrates the practical use of basic Python programming concepts. The project helped in understanding loops, user input, string manipulation, and Python's built-in random and string modules.

Through this project, I gained hands-on experience in developing a simple command-line application and learned how different Python concepts can be combined to create a useful program..

GitHub Repository:

[AmimaBatoool/Task-3 AmimaBatoool: A Python-based Random Password Generator that creates strong and secure passwords using letters, numbers, and special characters.](#)

Project status: Completed.